

A List Game

CPU TIME LIMIT

1 second

MEMORY LIMIT

1024 MB

DIFFICULTY

easy (2.2)

You are playing the following simple game with a friend:

1. The first player picks a positive integer X .
2. The second player gives a list of k positive integers $Y_1, \dots, Y_k$ such that $(Y_1 + 1)(Y_2 + 1) \cdots (Y_k + 1) = X$, and gets k points.

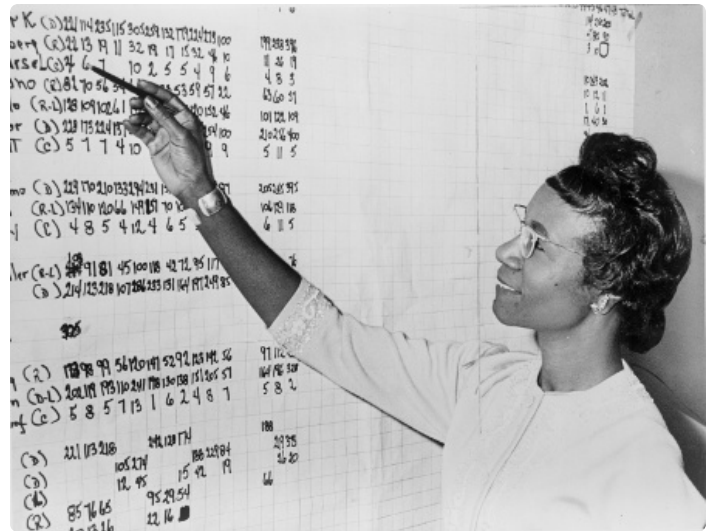
Write a program that plays the second player.

Input

The input consists of a single integer X satisfying $10^3 \leq X \leq 10^9$, giving the number picked by the first player.

Output

Write a single integer k , giving the number of points obtained by the second player, assuming she plays as well as possible.



Shirley Chisholm examining political statistics in 1965 (from Wikipedia Commons).

Sample Input 1

65536

Sample Output 1

16



Sample Input 2

127381

Sample Output 2

3
